



Economic policy uncertainty and narrative R&D disclosures

Keumah Jung^a, Sehee Kim^{b,*}

^a Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea

^b Chung-Ang University, 84, Heukseok-ro, Dongjak-gu, Seoul, Republic of Korea

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ABSTRACT

This study examines the impact of economic policy uncertainty on narrative R&D disclosure. Analyzing 10-K reports from 1997 to 2020, we find that the quantity of R&D disclosure increases with economic policy uncertainty. We also document that firms provide more forward-looking R&D disclosure during uncertain periods. Finally, we find that such impact is more pronounced for the firms with high return volatility and firms in the early-stage of business cycle. We contribute to the literature by showing how managers adjust R&D disclosures in response to economic policy uncertainty.

1. Introduction

Investors face greater uncertainty about a firm's value when macroeconomic uncertainty is high. Prior literature shows that managers respond to increased information asymmetry by providing more frequent voluntary disclosure (Nagar et al., 2019; Boone et al., 2021) and increasing the amount of information disclosed in annual reports (Jiang et al., 2022). These findings are consistent with the notion that managers have incentives to disclose more information to reduce negative capital market consequences.

In this paper, we extend prior literature by investigating the impact of economic policy uncertainty (EPU) on narrative Research and Development (R&D hereafter) disclosures. We focus specifically on textual R&D disclosures for several reasons. First, R&D activities are risky and long-term in nature, making R&D activity more vulnerable to macroeconomic uncertainty. Second, all R&D expenditures should be expensed under U.S. GAAP, making investors difficult to assess the value of R&D activities. Accordingly, the value-relevance of narrative R&D disclosure can be greater than that of other types of information during uncertain periods. Third, R&D disclosure incurs proprietary costs, which reduce net benefits of additional disclosure. Thus, it is an empirical question how managers adjust R&D disclosures in the periods of high uncertainty.

To measure economic policy uncertainty, we use the EPU index developed by Baker et al. (2016). Baker et al. (2016) construct the EPU index based on three components: news coverage of policy economic uncertainty, the number of federal tax code provisions set to expire, and forecaster disagreement over inflation and government purchases in the near future. They provide the monthly index, which allows us to examine time-series effects of policy uncertainty. To empirically measure narrative R&D disclosures, we examine the textual disclosure of 10-Ks from 1997 to 2020. Employing Merkley (2014)'s approach, we identify R&D-related sentences and quantify the R&D textual disclosures. Our findings suggest that managers tend to provide a greater quantity of R&D disclosure during the periods of high uncertainty. We also find that R&D related sentences include more forward-looking words in uncertain periods. Lastly, we find that such effects are stronger for firms with higher information demand, measured as stock return volatility or the

* Corresponding author at: School of Business Administration, Chung-Ang University, 84, Heukseok-ro, Dongjak-gu, Seoul, (06974) Republic of Korea.

E-mail addresses: kajung@snu.ac.kr (K. Jung), seheekim@cau.ac.kr (S. Kim).

early-stage of business cycle.

This paper contributes to the literature in two ways. First, we contribute to the literature that examines the impact of policy uncertainty on corporate disclosure. We extend Jiang et al. (2022)'s findings by focusing on narrative R&D disclosure in annual reports. While Jiang et al. (2022) examine the contents of entire annual reports, it is ex-ante unclear how managers determine R&D disclosure during uncertain periods because of the unique features of R&D activities. Second, our findings contribute to the literature on R&D disclosure. Prior studies mainly focus on firm or industry-level information demands such as earnings performance (Merkley, 2014), market competition (Cao et al., 2018), and brokerage house mergers and closures (Baruffaldi et al., 2024). Our study adds to the literature by focusing on macro-level information demand (i.e., economic policy uncertainty).

The remainder of the paper is organized as follows. Section 2 discusses prior literature and hypotheses. Section 3 describes the methodology. Section 4 reports the empirical findings. Section 5 concludes the paper.

2. Prior literature and hypothesis development

2.1. Prior literature on economic policy uncertainty

Prior studies examine the role of policy uncertainty in various aspects of corporate policies. A stream of literature documents that policy uncertainty affects firms' real activities. For instance, several studies find a negative association between policy uncertainty and capital investment (Julio and Yook, 2012; Gulen and Ion, 2016; Jens, 2017). According to real options theory, these results suggest that firms delay investment decisions until some uncertainty is resolved. Relatedly, Bonaime et al. (2018) find that merger and acquisition activities decline during periods of high policy uncertainty. In contrast, firms tend to increase their R&D investment as policy uncertainty increases, since the value of R&D growth options increases with uncertainty (de la Horra et al., 2022; Vo and Le, 2017). In terms of financing decisions, policy uncertainty is negatively associated with equity financing (Colak et al., 2017), while it is positively associated with the level of cash holding (Julio and Yook, 2012).

Another stream of literature investigates whether policy uncertainty influences disclosure. Specifically, Nagar et al. (2019) and Boone et al. (2021) find that policy uncertainty increases information asymmetry, prompting managers to release more voluntary disclosures to mitigate the adverse effects of this asymmetry. Jiang et al. (2022), which is particularly relevant to our paper, find that the length of annual reports increases with policy uncertainty. They further find that investors favorably value the increase in disclosure, resulting in lower equity financing costs.

2.2. Prior literature on R&D disclosure

U.S. GAAP requires all R&D expenditures to be expensed, making it challenging to convey the value of R&D activities solely through financial statements. Therefore, managers often use textual disclosures to provide additional information about R&D activities. Prior studies find that higher information asymmetry is associated with increased text-based R&D disclosure (Merkley 2014; Baruffaldi et al. 2024; Huang et al. 2021), with a focus on earnings performance (Merkley, 2014) and brokerage house mergers and closures (Baruffaldi et al., 2024). However, while information asymmetry encourages R&D disclosures, concerns about proprietary costs can discourage them, especially in highly competitive or R&D-intensive industries (Cao et al., 2018; Jones, 2007).

2.3. Hypotheses development

Based on the prior literature, two opposite predictions are plausible. On the one hand, consistent with the findings from Nagar et al. (2019) and Jiang et al. (2022), managers may be motivated to increase the quantity of R&D disclosure to reduce information asymmetry during uncertain periods. On the other hand, high macroeconomic uncertainty may incentivize managers to reduce R&D disclosures due to increased proprietary costs. Under greater uncertainty, firms are more likely to rely on disclosed R&D information from peers for strategic insights. Competitors can exploit R&D information to gain a competitive advantage. As a result, managers may withhold R&D disclosures to protect their competitiveness and prevent potential strategic disadvantages. Based on the discussion, it remains unclear whether and how managers adjust narrative R&D disclosures during periods of high uncertainty. Thus, we formulate our first hypothesis in a null form:

Hypothesis 1. *Economic Policy Uncertainty is not associated with narrative R&D disclosures.*

We further examine the content of narrative R&D disclosures during uncertain periods. Specifically, following Merkley (2014), we analyze forward-looking words in R&D-related sentences. On the one hand, managers may provide more value-relevant prospect information and thereby reduce information asymmetry. On the other hand, managers might provide opaque R&D disclosures that are less relevant to investors because managers are also uncertain about future prospect when macroeconomic uncertainty is high. Thus, we set our second hypothesis in a null form:

Hypothesis 2. *Economic Policy Uncertainty is not associated with forward-looking R&D disclosures*

Lastly, we investigate whether the relation between policy uncertainty and R&D disclosures varies with firm-specific information demand. Information demand generally increases with market volatility (Vlastakis and Markellos, 2012), and firms in the early-stage of business cycle are more concerned about information asymmetry (Dickinson et al., 2018). Therefore, we expect policy uncertainty to have a greater impact on firms experiencing higher stock return volatility or those in early stages. However, firm-specific information

Table 1
Descriptive statistics.

Variables	N	Mean	Std	1P	Q1	Med	Q3	99P
$LnR\&D_{i,t}$	45,706	2.567	1.394	0.000	1.792	2.708	3.434	5.733
$LnR\&D_Fls_{i,t}$	41,131	1.993	1.303	0.000	1.099	1.946	2.639	5.580
EPU_Comp_{t-1}	45,706	4.629	0.260	4.266	4.381	4.657	4.790	5.149
EPU_News_{t-1}	45,706	4.707	0.296	4.205	4.392	4.727	4.978	5.240
$adjROA_{i,t-1}$	45,706	0.050	0.227	-1.040	0.000	0.094	0.166	0.469
$Size_{i,t-1}$	45,706	5.792	2.205	1.296	4.171	5.711	7.268	11.473
$Age_{i,t-1}$	45,706	2.478	0.905	0.632	1.799	2.534	3.146	4.410
$SqAge_{i,t-1}$	45,706	4.727	1.985	0.576	3.275	4.911	6.205	8.795
$R\&DInt_{i,t-1}$	45,706	0.152	0.212	0.000	0.009	0.064	0.199	0.884
$SqR\&DInt_{i,t-1}$	45,706	0.068	0.160	0.000	0.000	0.004	0.040	0.782
$BtM_{i,t-1}$	45,706	0.529	0.543	-0.577	0.208	0.391	0.687	3.151
$CapInt_{i,t-1}$	45,706	0.323	0.232	0.003	0.125	0.288	0.484	0.895
$RetVol_{i,t-1}$	45,706	0.163	0.104	0.037	0.091	0.136	0.203	0.603
$adjROAVol_{i,t-1}$	45,706	0.079	0.118	0.002	0.018	0.038	0.087	0.748
$Lev_{i,t-1}$	45,706	0.185	0.209	0.000	0.002	0.122	0.296	0.971
$Issue_{i,t-1}$	45,706	0.769	0.422	0.000	1.000	1.000	1.000	1.000
$LnNonR\&D_{i,t}$	45,706	7.838	0.726	6.368	7.383	7.717	8.133	10.020
$LnNonR\&D_Fls_{i,t}$	41,131	6.018	0.588	4.382	5.666	6.087	6.430	7.223
$Intro_{i,t-1}$	45,706	0.186	0.389	0.000	0.000	0.000	0.000	1.000

Notes: All continuous variables are winsorized at the top and bottom one percentile. Please refer to [Appendix A1](#) for variable definitions.

demand may not play a significant role if managers are cautious with R&D disclosures to protect their competitiveness under increased macro-level uncertainty. Therefore, we propose our third hypothesis as follows:

Hypothesis 3a. *Stock return volatility does not moderate the association between Economic Policy Uncertainty and narrative R&D disclosures.*

Hypothesis 3b. *Business cycle does not moderate the association between Economic Policy Uncertainty and narrative R&D disclosures.*

3. Sample and research design

3.1. Sample and data

Our initial sample consists of U.S. firms from the Compustat, CRSP, and SEC EDGAR databases between 1995 and 2020. The sample period starts in 1995 because SEC EDGAR began in 1994, and one-year lag of economic policy uncertainty (EPU) data is required. We obtain textual R&D disclosures from EDGAR 10-K filings and the U.S. EPU index from a website operated by Baker, Bloom, and Davis.¹ We exclude financial firms (SIC codes 6000–6999) and utility firms (SIC codes 4900–4999). We also remove observations with missing information. The final sample consists of 45,706 firm-years from 1997 to 2020.

3.2. Research design

To test our hypotheses, we regress current-year R&D disclosure on prior-year EPU based on the [Eq. \(1\)](#) following prior studies (e.g., [Merkley, 2014](#); [Nagar et al., 2019](#); [Jiang et al., 2022](#)).

$$R\&D_{i,t} = \beta_0 + \beta_1 EPU_{t-1} + \beta_k \sum Controls_{i,t-1} + YearFE + FirmFE + \varepsilon_{i,t} \quad (1)$$

The dependent variable, $R\&D_{i,t}$, represents the quantity of R&D disclosure. We measure this as the natural logarithm of one plus the number of R&D-related sentences ($LnR\&D_{i,t}$) or forward-looking R&D sentences ($LnR\&D_Fls_{i,t}$) in 10-K filings for firm i in year t . We identify R&D sentences by looking for R&D keywords.² We then classify these R&D sentences as forward-looking if they include future tense words as in [Merkley \(2014\)](#).³ Our variable of interest is EPU_{t-1} , either EPU_Comp_{t-1} or EPU_News_{t-1} , which is the natural logarithm of three components-based or news-based economic policy uncertainty index for the U.S. in year $t-1$. The coefficient (β_1) captures the

¹ We obtain EPU data from <https://www.policyuncertainty.com/>. We thank Baker, Bloom, and Davis for generously sharing the data.

² The R&D keywords include terms such as “R&D,” “research and development,” “product development,” “technology development,” “clinical candidate,” “research collaboration,” “new patent,” and “applied for patent.” ([Merkley 2014](#)).

³ The forward-looking keywords include terms such as “will,” “could,” “should,” “expect,” “anticipate,” “plan,” “believe,” “may,” and “forecast” ([Merkley 2014](#)).

Table 2
Economic policy uncertainty and R&D disclosure.

Dep. Variable =	$LnR\&D_{i,t}$		$LnR\&D_Fls_{i,t}$	
	(1)	(2)	(3)	(4)
<i>EPU_Comp</i>_{<i>i,t-1</i>}	0.091** (2.51)		0.068* (1.67)	
<i>EPU_News</i>_{<i>i,t-1</i>}		0.066*** (2.97)		0.052** (2.02)
<i>adjROA</i> _{<i>i,t-1</i>}	0.004 (0.18)	0.004 (0.18)	-0.011 (-0.42)	-0.011 (-0.42)
<i>Size</i> _{<i>i,t-1</i>}	0.063*** (10.22)	0.063*** (10.22)	0.054*** (7.66)	0.054*** (7.66)
<i>Age</i> _{<i>i,t-1</i>}	0.219 (0.45)	0.219 (0.45)	0.679 (1.61)	0.679 (1.61)
<i>SqAge</i> _{<i>i,t-1</i>}	-0.117 (-0.56)	-0.117 (-0.56)	-0.324* (-1.79)	-0.324* (-1.79)
<i>R&DInt</i> _{<i>i,t-1</i>}	2.095*** (13.47)	2.094*** (13.47)	1.990*** (11.89)	1.990*** (11.88)
<i>SqR&DInt</i> _{<i>i,t-1</i>}	-1.698*** (-11.36)	-1.698*** (-11.36)	-1.656*** (-10.11)	-1.656*** (-10.10)
<i>BtM</i> _{<i>i,t-1</i>}	0.038*** (4.44)	0.038*** (4.44)	0.024** (2.42)	0.024** (2.42)
<i>CapInt</i> _{<i>i,t-1</i>}	0.009 (0.18)	0.009 (0.17)	-0.031 (-0.57)	-0.031 (-0.57)
<i>RetVol</i> _{<i>i,t-1</i>}	0.096*** (3.14)	0.096*** (3.14)	0.059 (1.64)	0.059 (1.63)
<i>adjROAVol</i> _{<i>i,t-1</i>}	0.006 (0.17)	0.006 (0.16)	0.020 (0.44)	0.020 (0.44)
<i>Lev</i> _{<i>i,t-1</i>}	0.043 (1.53)	0.043 (1.53)	0.019 (0.62)	0.019 (0.62)
<i>Issue</i> _{<i>i,t-1</i>}	0.029*** (4.10)	0.029*** (4.11)	0.030*** (3.73)	0.030*** (3.74)
<i>LnNonR&D</i> _{<i>i,t</i>}	0.102*** (16.47)	0.102*** (16.47)		
<i>LnNonR&D_Fls</i> _{<i>i,t</i>}			0.342*** (25.67)	0.342*** (25.67)
<i>Constant</i>	0.717** (2.43)	0.830*** (3.17)	-1.099*** (-3.70)	-1.029*** (-4.02)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Cluster	Firm	Firm	Firm	Firm
Observations	45,706	45,706	41,131	41,131
Adjusted R ²	0.931	0.931	0.901	0.901

Notes: This table reports the association between EPU and narrative R&D disclosure. The dependent variable is natural log of one plus the number of R&D-related sentences in 10-Ks in columns (1) and (2), while the dependent variable is natural log of one plus the number of R&D-related sentences that contain forward-looking words in columns (3) and (4). *, **, *** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively (two-tailed). All continuous variables are winsorized at the top and bottom one percentile. *t*-statistics based on standard errors clustered by firm are shown in parentheses. Please refer to [Appendix A1](#) for variable definitions.

association between economic policy uncertainty and the quantity of narrative R&D disclosure. *Controls*_{*i,t-1*} represents the control variables.⁴ Year and firm fixed effects are included to account for unobservable time and firm specific factors, respectively. Standard errors are clustered at the firm level. All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the effects of outliers. Detailed definitions of all variables are provided in [Appendix A1, Table A1](#).

4. Empirical findings

4.1. Descriptive statistics

[Table 1](#) represents descriptive statistics for key variables in our analyses. On average, firms disclose 30.96 R&D-related sentences in their annual reports, including 19.80 forward-looking sentences. The mean values for the three-component-based and news-based *EPU* are 4.629 and 4.707, respectively. In addition, when we estimate Pearson correlation coefficients among the variables in [Eq. \(1\)](#), *EPU* is positively correlated with the quantity of narrative R&D disclosure (untabulated).

⁴ We control for firm characteristics, including earnings (*adjROA*), size (*Size*), firm age (*Age*, *SqAge*), R&D investment (*R&DInt*, *SqR&DInt*), capital investment (*CapInt*), growth opportunity (*BtM*), volatility (*RetVol*, *adjROAVol*), risk (*Lev*), and equity issuance (*Issue*). We also control for the length of annual report (*NonR&D*) by measuring the amount of non-R&D (*LnNonR&D*) or forward-looking (*LnNonR&D_Fls*) information in 10-Ks.

Table 3
Cross-sectional analyses.

Panel A: Return volatility				
Dep. Variable =	$LnR\&D_{i,t}$		$LnR\&D_Fls_{i,t}$	
	(1)	(2)	(3)	(4)
$EPU_Comp_{i,t-1}$	0.010 (0.25)		-0.022 (-0.47)	
$EPU_Comp_{i,t-1} * RetVol_{i,t-1}$	0.486*** (4.54)		0.530*** (4.27)	
$EPU_News_{i,t-1}$		-0.025 (-0.92)		-0.034 (-1.06)
$EPU_News_{i,t-1} * RetVol_{i,t-1}$		0.517*** (5.74)		0.482*** (4.58)
$RetVol_{i,t-1}$	-2.157*** (-4.31)	-2.351*** (-5.47)	-2.402*** (-4.13)	-2.224*** (-4.43)
$adjROA_{i,t-1}$	0.002 (0.09)	0.003 (0.14)	-0.014 (-0.51)	-0.013 (-0.47)
$Size_{i,t-1}$	0.064*** (10.45)	0.065*** (10.53)	0.056*** (7.90)	0.056*** (7.92)
$Age_{i,t-1}$	0.212 (0.44)	0.204 (0.42)	0.662 (1.58)	0.656 (1.57)
$SqAge_{i,t-1}$	-0.116 (-0.56)	-0.113 (-0.55)	-0.319* (-1.78)	-0.316* (-1.77)
$R\&DInt_{i,t-1}$	2.079*** (13.39)	2.076*** (13.37)	1.973*** (11.82)	1.973*** (11.82)
$SqR\&DInt_{i,t-1}$	-1.679*** (-11.25)	-1.676*** (-11.24)	-1.635*** (-10.01)	-1.635*** (-10.02)
$BtM_{i,t-1}$	0.037*** (4.42)	0.037*** (4.38)	0.024** (2.40)	0.024** (2.36)
$CapInt_{i,t-1}$	0.005 (0.10)	0.004 (0.09)	-0.036 (-0.66)	-0.036 (-0.66)
$adjROAVol_{i,t-1}$	0.042 (1.51)	0.043 (1.53)	0.018 (0.40)	0.018 (0.40)
$Lev_{i,t-1}$	0.029*** (4.15)	0.030*** (4.20)	0.018 (0.60)	0.019 (0.62)
$Issue_{i,t-1}$	0.103*** (16.51)	0.103*** (16.54)	0.031*** (3.80)	0.031*** (3.82)
$LnNonR\&D_{i,t}$	-2.157*** (-4.31)	-2.351*** (-5.47)		
$LnNonR\&D_Fls_{i,t}$			-0.670** (-2.18)	-0.605** (-2.28)
Constant	1.096*** (3.56)	1.265*** (4.61)	-0.670** (-2.18)	-0.605** (-2.28)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Cluster	Firm	Firm	Firm	Firm
Observations	45,706	45,706	41,131	41,131
Adjusted R ²	0.931	0.931	0.901	0.901
Panel B: Business cycle				
Dep. Variable =	$LnR\&D_{i,t}$		$LnR\&D_Fls_{i,t}$	
	(1)	(2)	(3)	(4)
$EPU_Comp_{i,t-1}$	0.070* (1.90)		0.042 (1.03)	
$EPU_Comp_{i,t-1} * Intro_{i,t-1}$	0.130*** (5.62)		0.150*** (5.36)	
$EPU_News_{i,t-1}$		0.047** (2.11)		0.033 (1.24)
$EPU_News_{i,t-1} * Intro_{i,t-1}$		0.113*** (5.68)		0.116*** (4.79)
$Intro_{i,t-1}$	-0.601*** (-5.61)	-0.533*** (-5.68)	-0.695*** (-5.35)	-0.548*** (-4.79)
$adjROA_{i,t-1}$	0.002 (0.10)	0.002 (0.10)	-0.014 (-0.51)	-0.013 (-0.50)
$Size_{i,t-1}$	0.063*** (10.32)	0.063*** (10.33)	0.055*** (7.79)	0.055*** (7.78)
$Age_{i,t-1}$	0.220 (0.45)	0.219 (0.45)	0.670 (1.60)	0.671 (1.60)
$SqAge_{i,t-1}$	-0.118 (-0.57)	-0.118 (-0.57)	-0.322* (-1.79)	-0.322* (-1.80)

(continued on next page)

Table 3 (continued)

Panel A: Return volatility				
Dep. Variable =	$LnR\&D_{i,t}$		$LnR\&D_Fls_{i,t}$	
$R\&DInt_{i,t-1}$	2.094*** (13.51)	2.096*** (13.53)	1.990*** (11.96)	1.992*** (11.96)
$SqR\&DInt_{i,t-1}$	-1.693*** (-11.37)	-1.695*** (-11.38)	-1.650*** (-10.13)	-1.653*** (-10.13)
$BtM_{i,t-1}$	0.038*** (4.53)	0.038*** (4.51)	0.025** (2.52)	0.025** (2.48)
$CapInt_{i,t-1}$	0.007 (0.15)	0.006 (0.13)	-0.033 (-0.62)	-0.034 (-0.63)
$RetVol_{i,t-1}$	0.097*** (3.18)	0.094*** (3.11)	0.060* (1.67)	0.057 (1.60)
$adjROAVol_{i,t-1}$	0.007 (0.18)	0.008 (0.22)	0.020 (0.45)	0.022 (0.49)
$Lev_{i,t-1}$	0.044 (1.57)	0.044 (1.57)	0.020 (0.65)	0.020 (0.65)
$Issue_{i,t-1}$	0.029*** (4.13)	0.029*** (4.13)	0.031*** (3.77)	0.031*** (3.76)
$LnNonR\&D_{i,t}$	0.102*** (16.48)	0.102*** (16.48)		
$LnNonR\&D_Fls_{i,t}$			0.342*** (25.73)	0.342*** (25.71)
Constant	0.820*** (2.78)	0.921*** (3.52)	-0.970*** (-3.28)	-0.926*** (-3.64)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Cluster	Firm	Firm	Firm	Firm
Observations	45,706	45,706	41,131	41,131
Adjusted R ²	0.931	0.931	0.901	0.901

Notes: This table examines whether the relation between EPU and narrative R&D disclosure varies with firm-level information demand. Panel A examines firm-level return volatility, whereas Panel B examines the stage of business cycle as a proxy for firm-level information demand. *, **, *** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively (two-tailed). All continuous variables are winsorized at the top and bottom one percentile. *t*-statistics based on standard errors clustered by firm are shown in parentheses. Please refer to [Appendix A1](#) for variable definitions.

4.2. EPU and narrative R&D disclosures

Table 2 reports the estimation results of Eq. (1). In columns (1)-(2), we examine the association between economic policy uncertainty and the quantity of narrative R&D disclosure, which is measured as $LnR\&D$. In addition, EPU_Comp (EPU_News) is used to measure economic policy uncertainty in column (1) (column (2)). In columns (1)-(2), we find positive and significant coefficients on EPU (column (1): coefficient = 0.091, *t*-stat. = 2.51; column (2): coefficient = 0.066, *t*-stat. = 2.97), suggesting that firms provide greater amount of narrative R&D disclosure in 10-K when policy uncertainty increases.⁵ The results reject hypothesis 1, which is stated in a null form, suggesting that managers are motivated to disclose more information during uncertain periods to reduce information asymmetry. This finding suggests that managers believe the benefits of increased R&D disclosures outweigh the costs, aligning with the findings of [Jiang et al. \(2022\)](#).⁶

In columns (3) and (4), we explore the content of R&D disclosures to determine whether firms provide more useful information to compensate for heightened policy uncertainty. We examine forward-looking information in these R&D disclosures, reducing the sample size to 41,131 firm-year observations. We find that the forward-looking R&D disclosure is greater when policy uncertainty increases (column (3): coefficient = 0.068, *t*-stat. = 1.67; column (4): coefficient = 0.052, *t*-stat. = 2.02), showing that 10-Ks include more forward-looking R&D disclosures during uncertain periods.⁷ Thus, hypothesis 2 is also rejected, indicating that managers tend to provide more informative R&D disclosure when policy uncertainty increases. In sum, the results in Table 2 suggest that managers respond to high policy uncertainty by providing a greater quantity of R&D-related sentences and forward-looking details to reduce information asymmetry.

⁵ Specifically, with a 1% increase in EPU, we would expect R&D disclosure to increase by 0.09%. Similarly, 1% increase in EPU based on news leads to a 0.07% increase in R&D disclosure.

⁶ According to real options theory, firms increase R&D investment as policy uncertainty rises ([de la Horra et al., 2022](#); [Vo and Le, 2017](#)). R&D disclosure may reflect this increased investment. To account for this, we control for R&D investment and find a positive relation between policy uncertainty and R&D disclosure.

⁷ 1% increase in EPU (news-based EPU) corresponds to a 0.07% (0.05%) increase in forward-looking R&D disclosure.

4.3. Cross-sectional analyses

In this section, we test hypotheses 3a and 3b. First, we test hypothesis 3a by re-estimating Eq. (1), incorporating the previous year's stock return volatility (*RetVol*) and the interaction term between *RetVol* and *EPU*. Second, to test hypothesis 3b, we categorize firms into five stages following Dickinson (2011),⁸ and identify early-stage firms as those in the introduction phases (*Intro*).

Panel A of Table 3 shows that the positive effect of policy uncertainty on R&D disclosure is indeed greater for firms with high return volatility. Specifically, these firms are likely to provide more narrative R&D disclosures, which include more forward-looking words. Panel B of Table 3 shows that our main findings are more pronounced for the firms in the early-stage of business cycle. For the firms in the early-stage, the quantity of R&D disclosures is greater, and R&D disclosures contain more forward-looking narratives.

Overall, the results in Table 3 reject the null hypotheses of 3a and 3b, indicating that firms disclose more R&D information, particularly prospective information, in response to higher macro- and firm-level uncertainty. This suggests that the managerial intention to mitigate information asymmetry drives firms to provide more qualitative R&D information under conditions of greater uncertainty.

5. Conclusion

In this paper, we examine how firms provide narrative R&D disclosures under high economic policy uncertainty. Identifying R&D-related sentences in 10-K reports, we find that firms provide greater quantity of R&D disclosure that contains more forward-looking words during uncertain periods. We further find that the positive relation between economic policy uncertainty and R&D disclosure is stronger for firms with high information demand, measured as return volatility and early-stage of business cycle. We extend prior studies by focusing on R&D disclosures. Given that we examine disclosure choice, we look forward to future research that addresses capital market consequences of R&D disclosure under uncertainty.

CRedit authorship contribution statement

Keumah Jung: Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization, Investigation, Software. **Sehee Kim:** Writing – review & editing, Writing – original draft, Project administration, Conceptualization, Supervision.

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Appendix A1

Table A1

Variable definitions.

Variable	Definition
<i>LnR&D</i>	= Natural log of (1 + total number of R&D-related sentences in 10-K) following Merkley (2014);
<i>LnR&D_Fls</i>	= Natural log of (1 + total number of R&D-related sentences containing forward-looking words) following Merkley (2014);
<i>EPU_Comp</i>	= Natural log of the average U.S. Policy Uncertainty Index (three-component index) for the fiscal year;
<i>EPU_News</i>	= Natural log of the average U.S. Policy Uncertainty Index (news-based index) for the fiscal year;
<i>adjROA</i>	= Operating income before R&D expense, scaled by total assets;
<i>Size</i>	= Firm size, measured as natural log of market value of equity;
<i>Age</i>	= Firm age, calculated as natural log of (1 + number of years a firm has been listed in the CRSP database);
<i>SqAge</i>	= Square of <i>Age</i> ;
<i>R&DInt</i>	= R&D expense divided by total operating expense;
<i>SqR&DInt</i>	= Square of <i>R&DInt</i> ;
<i>BtM</i>	= Ratio of book value to market value at the end of the fiscal period;
<i>CapInt</i>	= Sum of property, plant, and equipment (PP&E) and inventories, divided by total assets;
<i>RetVol</i>	= Standard deviation of monthly returns during the fiscal period;
<i>adjROAVol</i>	= Standard deviation of adjusted return on assets (<i>adjROA</i>) over the recent three years;
<i>Lev</i>	= Total debt divided by total assets;
<i>Issue</i>	= Indicator variable equal to 1 if the firm had a positive net stock issuance during the fiscal year, following Baker and Xuan (JFE 2016), and 0 otherwise;
<i>LnNonR&D</i>	= Natural log of (1 + total number of non-R&D-related sentences in 10-K);
<i>LnNonR&D_Fls</i>	= Natural log of (1 + total number of non-R&D-related sentences containing forward-looking words);
<i>Intro</i>	= Indicator variable equal to 1 if the firm is in the introduction stage of the business cycle, following Dickinson (2011), and 0 otherwise.

⁸ The business cycle includes five stages: introduction, growth, maturity, shakeout, and decline (Dickinson, 2011).

Data availability

Data will be made available on request.

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